Yunbei Zhang

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Research Interests

- **Test/Inference-Time Learning:** Improving reasoning via RL with unverified rewards, and adapting to new and changing domains.
- **Safety & Alignment:** Ensuring frontier agents are robust, reliable, and aligned with human values.
- **Efficient Model Re-use:** Lightweight reprogramming and prompting for adapting pre-trained models.

Education

- **Tulane University** — *Aug 2022 - Present*
  Ph.D. in Computer Science (Advisor: Prof. Jihun Hamm) — New Orleans, LA, USA
  - Focus: Developing efficient, robust, and generalizable ML methods for real-world deployment.
- **Osaka University** — *Apr 2018 - Mar 2022*
  B.S. in Mathematics (Award of Excellence) — Osaka, Japan
  - Honors: Osaka University Award of Excellence, Multiple Merit-based Scholarships
- **Peking University** — *Jul 2013 - May 2015*
  Incomplete — Beijing, China
  - Withdrew after two years to pursue alternative academic and career opportunities.

Professional Experience

- **Oak Ridge National Laboratory** — *Jan 2026 - April 2026 (Expected)*
  Research Intern — Oak Ridge, TN, USA
  - Uncertainty Quantification for Large Foundation Model and Multi-modal LLMs
- **Amazon** — *Sep 2025 - Dec 2025*
  Applied Scientist Intern — Seattle, WA, USA
  - Multi-modal LLM RL-based post-training and Safety
  - Automatic Multimodal Red Teaming via Agentic Planning
- **KLA Corporation** — *May 2025 - Aug 2025*
  Machine Learning Research Intern — Milpitas, CA, USA
  - Applied VFM Knowledge Distillation for compact Object Detection
- **Tulane University** — *Jan 2024 - May 2024*
  Teaching Assistant (**Outstanding TA Award 2024**) — New Orleans, LA, USA
  - Received Outstanding Teaching Assistant Award for exceptional student mentorship

Selected Publications (*equal contribution)

Conference Papers:

[ICML'25] DPCore: Dynamic Prompt Coreset for Continual Test-Time Adaptation
Y. Zhang, A. Mehra, S. Niu, J. Hamm

[NeurIPS'25] Doctor Approved: Generating Medically Accurate Skin Disease Images through AI-Expert Feedback
J. Wang*, **Y. Zhang***, Z. Ding, J. Hamm

[NeurIPS'24] Understanding the Transferability of Representations via Task-Relatedness
A. Mehra, **Y. Zhang**, J. Hamm

[TMLR] Prompt-based Adaptation in Large-scale Vision Models: A Survey
Y. Zhang*, X. Xiao*, L. Zhao*, et al.

[ICASSP'26] CTR-LoRA: Curvature-Aware and Trust-Region Guided Low-Rank Adaptation for Large Language Models
Z. Wang, M. Mo, X. Xiao, C. Liu, C. Ma, **Y. Zhang**, et al.

[USENIX'25] SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy
Y. Cheung, **Y. Zhang**, N. Marrouche, J. Hamm

[WACV'25] OT-VP: Optimal Transport-guided Visual Prompting for Test-Time Adaptation
Y. Zhang, A. Mehra, J. Hamm **[Oral Presentation]**

[ACM MM'25] Visual Instance-aware Prompt Tuning
Y. Zhang*, X. Xiao*, X. Li, T. Wang, X. Wang, Y. Wei, J. Hamm, M. Xu

[ACM MM'25] eSkinHealth: A Multimodal Dataset for Neglected Tropical Skin Diseases
J. Wang, X. Hu, **Y. Zhang**, et al.

[WACV'26] RoadBench: A Vision-Language Foundation Model for Road Damage Understanding
X. Xiao, **Y. Zhang**, J. Wang, et al.

[WACV'24] On the Fly Neural Style Smoothing for Risk-Averse Domain Generalization
A. Mehra, **Y. Zhang**, B. Kailkhura, J. Hamm

Workshop Papers:

- [ICMLW'25] Describe Anything in Medical Images
- [CVPRW'24] Achieving Reliable and Fair Skin Lesion Diagnosis via Unsupervised Domain Adaptation
- [CVPRW'24] Test-time Assessment of a Model's Performance on Unseen Domains via Optimal Transport

Recent Submissions (Under Review):

- Visual Exclusivity Attacks: Automatic Multimodal Red Teaming via **Agentic Planning**
- **Agents in the Wild**: Safety, Society, and the Illusion of Sociality on Moltbook
- Wan-R1: Verifiable-Reinforcement Learning for Generalizable Maze-Solving **Video Generation**
- Adapting in the Dark: Towards Stable and Efficient **Black-Box** Test-Time Adaptation
- Prime Once, then Reprogram Locally: Efficient Alternative to **Black-Box** Model Reprogramming
- Are **Medical Vision-Language Foundation Models** Ready for Dermatology
- Not All Directions Matter: Toward Structured and Task-Aware **Low-Rank Adaptation**
- Seeing Clearly, Reasoning Confidently: Plug-and-Play Remedies for **Vision Language Model** Blindness
- Continual Test-Time Adaptation: A Comprehensive Survey

Honors & Awards

- **Lambda's Research Grant** — *2025*
  Lambda
- **Outstanding Teaching Assistant Award** — *2024*
  Tulane University
- **Osaka University Award of Excellence** — *2019*
  Osaka University
- **Monbukagakusho Honors Scholarship** — *2018, 2021*
  Ministry of Education, Japan
- **School of Science Honor Scholarship (4 years)** — *2018-2021*
  Osaka University
- **Ichikawa International Scholarship** — *2021*
  Osaka University

Academic Service

- **Conference Reviewer:** NeurIPS 2025 **(Top Reviewer)**, ICLR 2025-2026, ACM MM 2025, CVPR 2026, ECCV 2026, ICML 2026
- **Journal Reviewer:** Transactions on Machine Learning Research
- **Volunteer:** NeurIPS 2023 Outreach Program

Technical Skills

- **Languages:** Python, C++, MATLAB, Bash, LaTeX
- **ML Frameworks:** PyTorch, TensorFlow, JAX, Hugging Face Transformers, scikit-learn
- **Tools:** Git, Docker, CUDA, Weights & Biases, AWS, Google Cloud Platform
- **Natural Languages:** English (Fluent), Japanese (Fluent), Chinese (Native)